Term 3 — Lesson 1 Practice Activity (C7) with Solutions

Contents

Evaluated criterion

- C7: Uses one-sample mean z-tests with known σ to find the minimum sample size required to reject H_0, then interprets the result in context.

1 Reference rule for all problems

For the sample mean case,

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Solving for n,

$$z = \frac{(\bar{x} - \mu_0)\sqrt{n}}{\sigma} \quad \longrightarrow \quad z\sigma = (\bar{x} - \mu_0)\sqrt{n} \quad \longrightarrow \quad \sqrt{n} = \frac{z\sigma}{\bar{x} - \mu_0} \quad \longrightarrow \quad n = \left(\frac{z\sigma}{\bar{x} - \mu_0}\right)^2.$$

Using a required critical value z^*, the full expression is

$$n = \left(\frac{z^*\sigma}{\bar{x} - \mu_0}\right)^2.$$

For the proportion case, the notation changes from $\bar{x}$ to $\hat{p}$, and from μ_0 to p_0. The standard deviation can be calculated directly using p_0:

$$\sigma_{\hat{p}} = \sqrt{p_0(1 - p_0)}.$$

Therefore, for the proportion case,

$$z = \frac{\hat{p} - p_0}{\frac{\sqrt{p_0(1-p_0)}}{\sqrt{n}}} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}..$$

Solving for n,

$$z = \frac{(\hat{p} - p_0)\sqrt{n}}{\sqrt{p_0(1-p_0)}} \qquad \longrightarrow \qquad \sqrt{n} = \frac{z\sqrt{p_0(1-p_0)}}{\hat{p} - p_0} \qquad \longrightarrow \qquad n = \left(\frac{z\sqrt{p_0(1-p_0)}}{\hat{p} - p_0}\right)^2.$$

Using a required critical value z^*, the full expression is

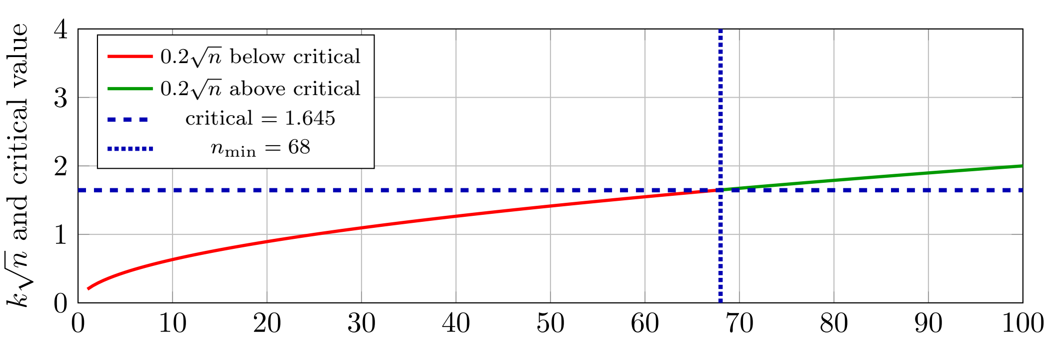
$$n = \frac{(z^*)^2 p_0(1-p_0)}{(\hat{p} - p_0)^2}.$$

Reject H_0 when: higher-tail $z \geq z_\alpha$, lower-tail $z \leq -z_\alpha$, two-tailed $|z| \geq z_{\alpha/2}$.

2 Problem 1: Right-tail, one significance level

C7

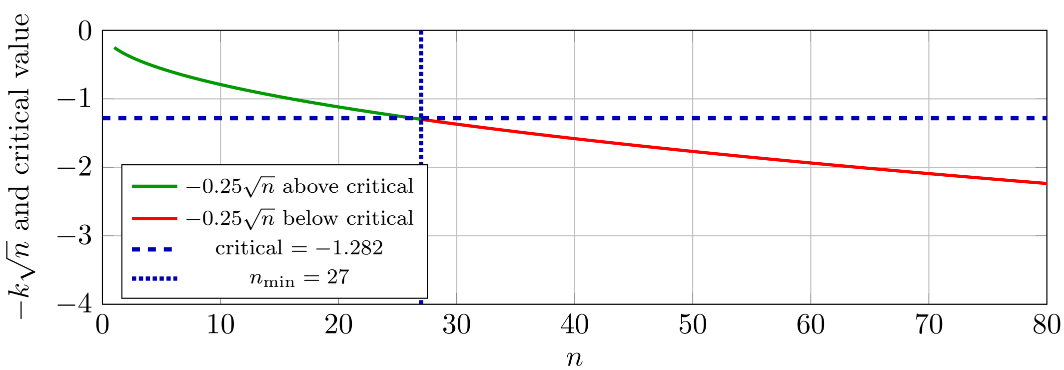
- **Context:** A previous study found that the average export price is 120. However, newly obtained data showed an observed export average of 124. We want to determine whether there is enough evidence to update the previous study's claim.
- **Parameter:** μ average export price.
- **Hypotheses:** $H_0 : \mu = 120$, $H_a : \mu > 120$.
- **Known standard deviation:** $\sigma = 20$, **Observed mean:** $\bar{x} = 124$, **Significance:** $\alpha = 5\%$.
- **Test statistic:** $z(n) = \frac{124-120}{20/\sqrt{n}} = 0.2\sqrt{n}$.
- **Critical value:** $z_{5\%} = 1.645$.
- **Inequality:** $0.2\sqrt{n} \geq 1.645 \Rightarrow n \geq 67.65$.
- **Minimum sample size:** $n_{\min} = 68$.
- **Interpretation:** Need at least 68 records to reject H_0 and conclude that there is enough evidence to update the previous study's claim.



3 Problem 2: Left-tail, one significance level

C7

- **Context:** A previous study found that the average delivery time is 18 days. However, newly obtained delivery-time records showed an observed average of 17 days. We want to determine whether there is enough evidence to update the previous study's claim and conclude that the average delivery time has decreased.
- **Parameter:** μ average delivery days.
- **Hypotheses:** $H_0 : \mu = 18$, $H_a : \mu < 18$.
- **Known standard deviation:** $\sigma = 4$, **Observed mean:** $\bar{x} = 17$, **Significance:** $\alpha = 10\%$.
- **Test statistic:** $z(n) = -0.25\sqrt{n}$.
- **Critical value:** $-z_{10\%} = -1.282$.
- **Inequality:** $-0.25\sqrt{n} \leq -1.282 \Rightarrow n \geq 26.30$.
- **Minimum sample size:** $n_{\min} = 27$.
- **Interpretation:** Need at least 27 observations to reject H_0 and conclude that there is enough evidence to update the previous study's claim.



4 Problem 3: Comparing a left-tail test and a two-tailed test

C7

- **Context:** A previous study found that the average income is 900. However, newly obtained survey data showed an observed average income of 885. Since the observed average is lower than 900, a left-tail test can be used if the goal is only to test whether income decreased. A two-tailed test can also be used if the goal is to determine whether the previous study's claim should be updated in either direction.
- **Test setup:** $\mu_0 = 900$, $\bar{x} = 885$, $\sigma = 60$, so

$$z = \frac{885 - 900}{60/\sqrt{n}} = -0.25\sqrt{n}.$$

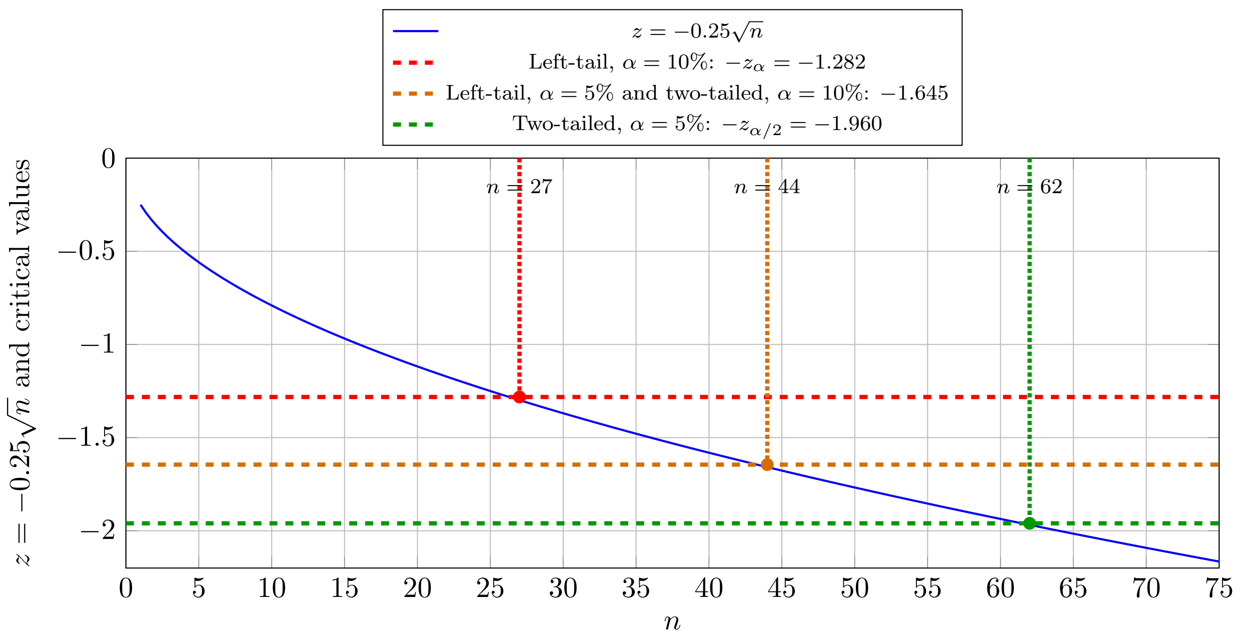
- **Left-tail test:** $H_0 : \mu = 900$, $H_a : \mu < 900$. Reject H_0 when $z \leq -z_\alpha$.

$$\alpha = 10\% \Rightarrow n_{\min} = 27, \qquad \alpha = 5\% \Rightarrow n_{\min} = 44.$$

- **Two-tailed test:** $H_0 : \mu = 900$, $H_a : \mu \neq 900$. Reject H_0 when $|z| \geq z_{\alpha/2}$.

$$\alpha = 10\% \Rightarrow n_{\min} = 44, \qquad \alpha = 5\% \Rightarrow n_{\min} = 62.$$

- **Comparison:** For the same significance level, the two-tailed test requires a larger sample size because it uses a larger critical value. In this context, the left-tail test only checks for a decrease, while the two-tailed test checks for any meaningful change from the previous study's value.



5 Problem 4: Comparing a right-tail test and a two-tailed test

C7

- **Context:** A previous study found that the average income is 900. However, newly obtained survey data showed an observed average income of 915. Since the observed average is higher than 900, a higher-tail test can be used if the goal is only to test whether income increased. A two-tailed test can also be used if the goal is to determine whether the previous study's claim should be updated in either direction.

- **Test setup:** $\mu_0 = 900$, $\bar{x} = 915$, $\sigma = 60$, so

$$z = \frac{915 - 900}{60/\sqrt{n}} = 0.25\sqrt{n}.$$

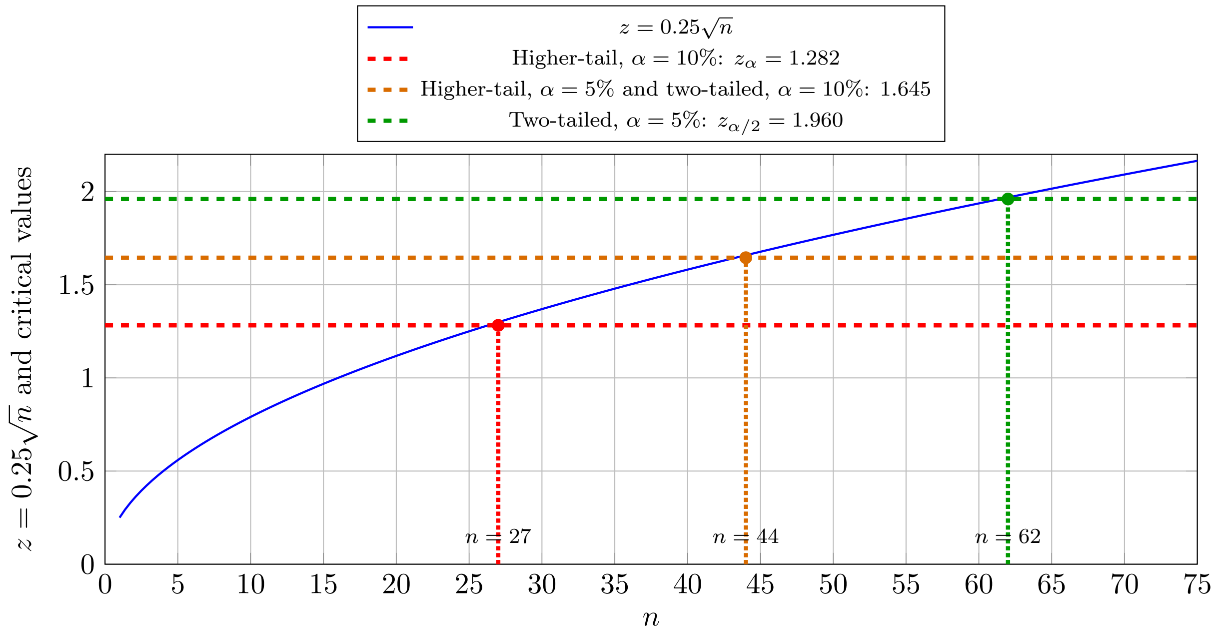
- **Higher-tail test:** $H_0 : \mu = 900$, $H_a : \mu > 900$. Reject H_0 when $z \geq z_\alpha$.

$$\alpha = 10\% \Rightarrow n_{\min} = 27, \qquad \alpha = 5\% \Rightarrow n_{\min} = 44.$$

- **Two-tailed test:** $H_0 : \mu = 900$, $H_a : \mu \neq 900$. Reject H_0 when $|z| \geq z_{\alpha/2}$.

$$\alpha = 10\% \Rightarrow n_{\min} = 44, \qquad \alpha = 5\% \Rightarrow n_{\min} = 62.$$

- **Comparison:** For the same significance level, the two-tailed test requires a larger sample size because it uses a larger critical value. In this context, the higher-tail test only checks for an increase, while the two-tailed test checks for any meaningful change from the previous study's value.



6 Problem 7: Left-tail test with multiple significance levels

C7

- **Context:** A previous study found that the average production index is 150. However, newly obtained data showed an observed average production index of 144. Since the observed average is lower than 150, a left-tail test is used to determine whether there is enough evidence to update the previous study's claim and conclude that the average production index has decreased.

- **Test setup:** $\mu_0 = 150$, $\bar{x} = 144$, $\sigma = 30$, so

$$z = \frac{144 - 150}{30/\sqrt{n}} = -0.2\sqrt{n}.$$

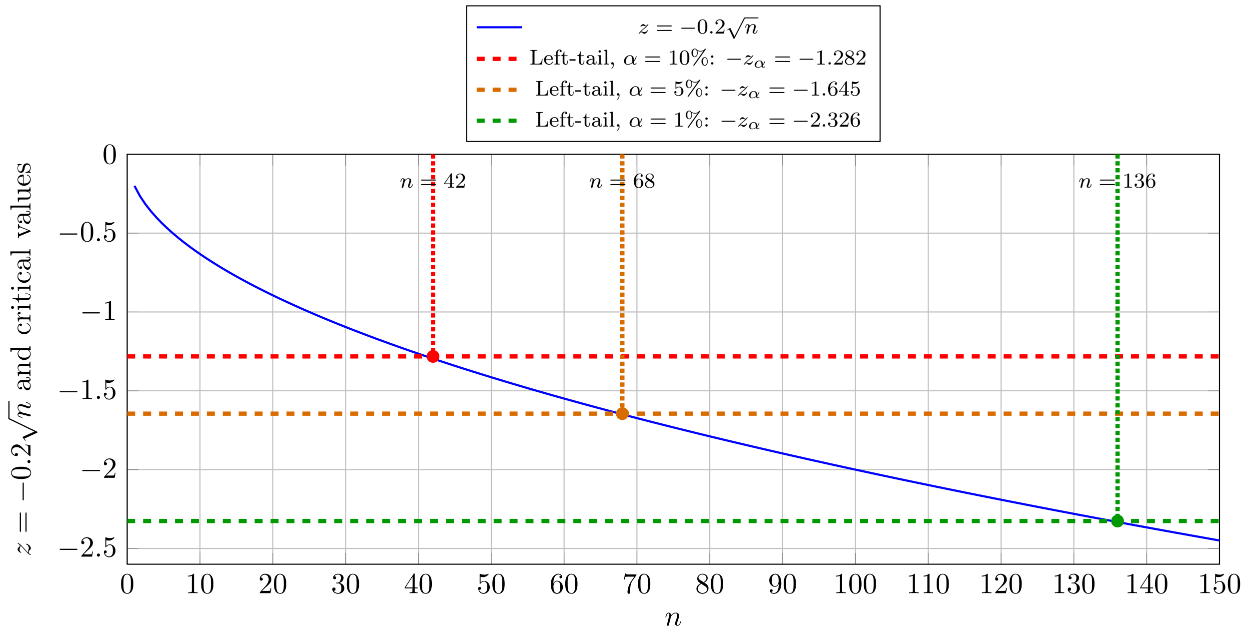
- **Left-tail test:** $H_0 : \mu = 150$, $H_a : \mu < 150$. Reject H_0 when $z \leq -z_\alpha$.

$$\alpha = 10\% \Rightarrow -z_\alpha = -1.282 \quad \longrightarrow \quad -0.2\sqrt{n} \leq -1.282 \quad \longrightarrow \quad n_{\min} = 42.$$

$$\alpha = 5\% \Rightarrow -z_\alpha = -1.645 \quad \longrightarrow \quad -0.2\sqrt{n} \leq -1.645 \quad \longrightarrow \quad n_{\min} = 68.$$

$$\alpha = 1\% \Rightarrow -z_\alpha = -2.326 \quad \longrightarrow \quad -0.2\sqrt{n} \leq -2.326 \quad \longrightarrow \quad n_{\min} = 136.$$

- **Comparison:** As α becomes smaller, the critical value moves farther into the left tail. Therefore, a stricter significance level requires a larger minimum sample size.



7 Problem 6: Right-tail test with multiple significance levels

C7

- **Context:** A previous study found that the average production index is 150. However, newly obtained data showed an observed average production index of 156. Since the observed average is higher than 150, a right-tail test is used to determine whether there is enough evidence to update the previous study's claim and conclude that the average production index has increased.
- **Test setup:** $\mu_0 = 150$, $\bar{x} = 156$, $\sigma = 30$, so

$$z = \frac{156 - 150}{30/\sqrt{n}} = 0.2\sqrt{n}.$$

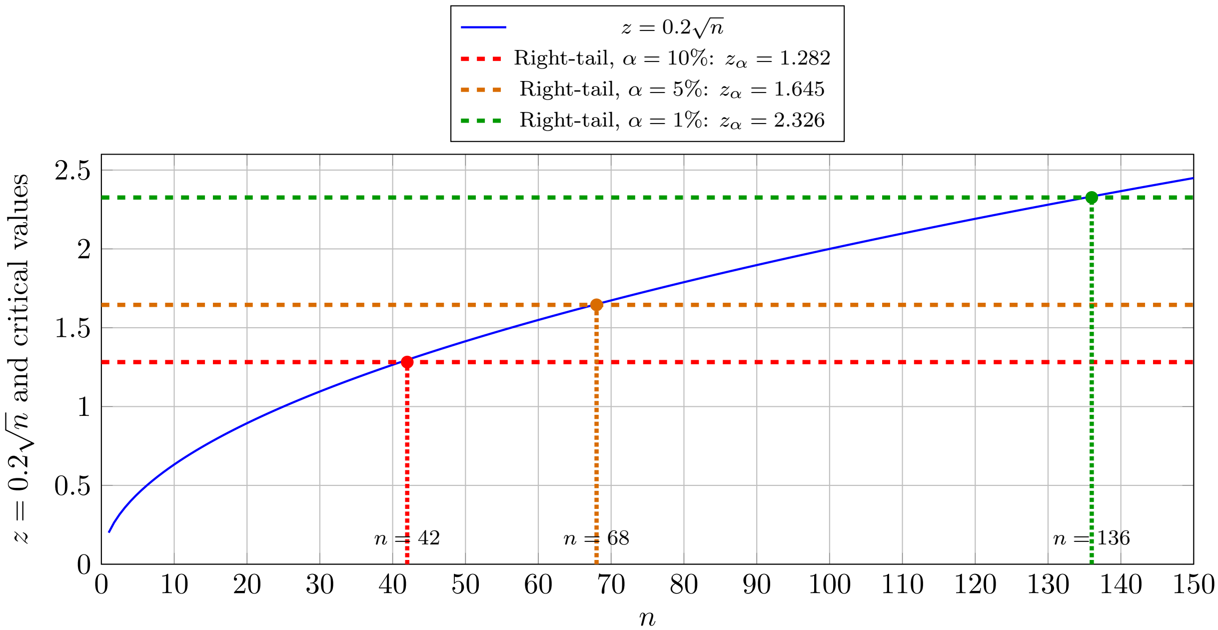
- **Right-tail test:** $H_0 : \mu = 150$, $H_a : \mu > 150$. Reject H_0 when $z \geq z_\alpha$.

$$\alpha = 10\% \Rightarrow z_\alpha = 1.282 \quad \longrightarrow \quad 0.2\sqrt{n} \geq 1.282 \quad \longrightarrow \quad n_{\min} = 42.$$

$$\alpha = 5\% \Rightarrow z_\alpha = 1.645 \quad \longrightarrow \quad 0.2\sqrt{n} \geq 1.645 \quad \longrightarrow \quad n_{\min} = 68.$$

$$\alpha = 1\% \Rightarrow z_\alpha = 2.326 \quad \longrightarrow \quad 0.2\sqrt{n} \geq 2.326 \quad \longrightarrow \quad n_{\min} = 136.$$

- **Comparison:** As α becomes smaller, the critical value becomes larger. Therefore, a stricter significance level requires a larger minimum sample size.



$z = 0.2\sqrt{n}$
Right-tail, $\alpha = 10\%$: $z_\alpha = 1.282$
Right-tail, $\alpha = 5\%$: $z_\alpha = 1.645$
Right-tail, $\alpha = 1\%$: $z_\alpha = 2.326$
$z = 0.2\sqrt{n}$ and critical values
$n = 42$
$n = 68$
$n = 136$
n